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CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

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ABSTRACT

After analysing the financial data from a European bank,checking the problems faced by banks in knowing about churn and retention. The main goal of this study is to help people to make good decisions and spot risks faster by analysing data properly instead of relying on their gut feeling. First, we took the raw CSV files and cleaned them up in Python. With NumPy and Pandas, we sorted out missing info, fixed inconsistencies, and made sure everything lined up right. Once the data was cleaned properly, then we started the main work. We checked how transactions worked at different levels—across departments, payroll categories, the whole structure. We created a set of KPIs to track what is the relationship between churn and customer behaviour. That only focusing on financial data cannot help to understand churn and retention. We followed different KPIs and we also did segmentation of employees and then analysed the churn percentage of each category. Then we check how improvement in customer behaviour influences retention. After that we created a streamlit dashboard which helps to understand and give a clear image of what is actually going on and what can be done in order to improve retention.

1. INTRODUCTION

Customer churn is one of the most critical challenges in the banking sector. It has a direct effect on profitability and long term customer relationships. As with the increasing competition and easy availability of financial services, customers change their banks quickly if their needs are not met. That's why understanding customer churn is not only important for profitability purposes but also for long term growth. There are many traditional studies which only focus on demographics and financial data of the organization, but I have found that understanding the customer behaviour is much more important than these during analyzing retention.

In this internship project, I have analyzed customer data and I found how engagement levels and product utilization impact churn. The purpose of this study is to create general, and data-driven insights which will help to get more effective retention strategies in banking sectors and will bridge the gap between data analytics and business decision making.

2. PROBLEM STATEMENT

Even after having large data sets of customers, banks are still facing problems many times about the main causes of customer churn. Most of the time retention strategies don't work where decision making fails, these strategies fail as these strategies are made on assumption or less analysis.

It is found that the main issue for this failure is, not focusing on customer behaviour, mainly engagement and product interaction. By analysing customers data financially only provides information about the customer value, it does not provide information about the relationship between customers and banks.

This study will help to bridge the gap by analyzing customer behaviour that how engagement and product interaction has effects on churn and how they influence retention. The main objective of this project is to provide a structured and data driven approach for enhancing retention strategies.

3. OBJECTIVES OF THE STUDY

The primary objective of this study is to analyze customer churn from a behavioral perspective and develop strategies to improve retention.

The specific objectives are:

- To understand how customer engagement affects churn, by comparing the behavior of active and inactive customers.
- To analyze whether customers who use more banking products tend to stay longer and show higher loyalty.
- To examine if having a higher account balance actually leads to better retention, or if such customers can still leave the bank.
- To identify different groups of customers based on their behavior, especially those who are more likely to churn.
- To explore patterns in customer data and use those insights to understand the main reasons behind customer exit.

- To suggest practical strategies that banks can use to improve customer retention and build stronger relationships with their customers.

4. DATASET DESCRIPTION

The dataset used in this project has customer-level information provided by European bank institutions which includes a combination of demographic, financial, and behavioral variables that will help in understanding customer characteristics.

The key variables are categorized as follows:

Demographic Variables:

- Geography (France, Germany, Spain)
- Gender
- Age

Financial Variables:

- Credit Score
- Account Balance
- Estimated Salary

Behavioral Variables:

- IsActiveMember (indicates customer engagement)
- NumOfProducts (number of banking products used)
- Tenure (duration of relationship with the bank)
- HasCrCard (credit card ownership)

Target Variable:

- Exited (1 = customer has churned, 0 = customer retained)

The dataset used in this project gives a strong base for analyzing how behavioral and financial factors influence customer churn.

5.METHODOLOGY

A proper framework was made so that the analysis is systematic, accurate, and aligned with the objectives of the project study.

5.1 Data Preparation

The first step done is about reviewing and validating the dataset to ensure data quality and consistency. Basic checks were executed in order to confirm that there were no major inconsistencies or missing values affecting the analysis.

To enhance the analytical depth, additional variables were created during study:

- **Relationship Score:** A combination metric based on customer engagement and product usage
- **High-Value Inactive Customers:** Customers with high balances but low engagement
- **Product Type:** Classification of customers into single-product and multi-product users.

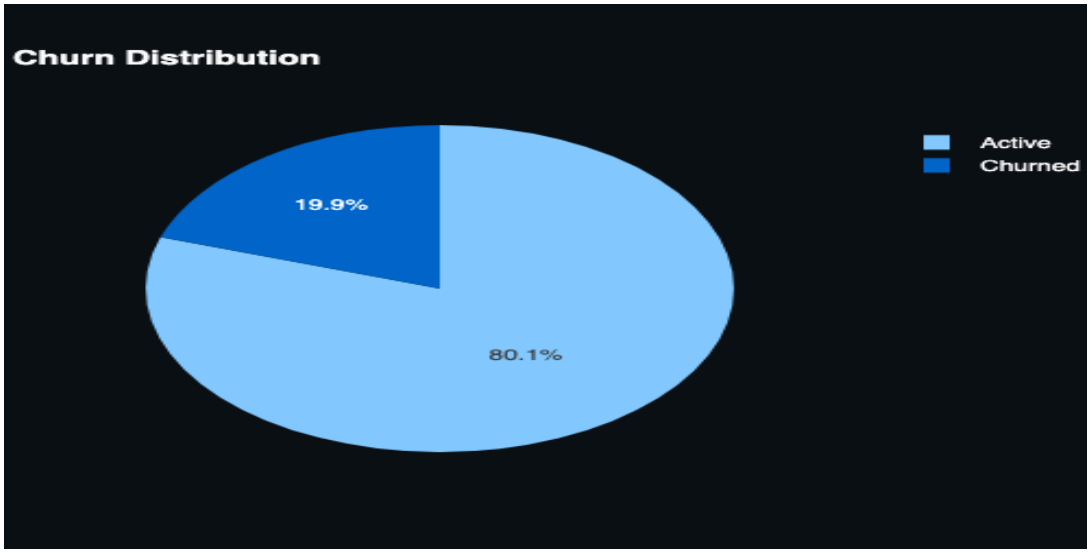
These derived variables helped in better understanding customer behavior.

5.2 Exploratory Data Analysis (EDA)

The exploratory data analysis was performed in order to understand patterns in the dataset and identify key factors influencing customer churn. The main observations are as follows:

5.2.1 Churn Distribution

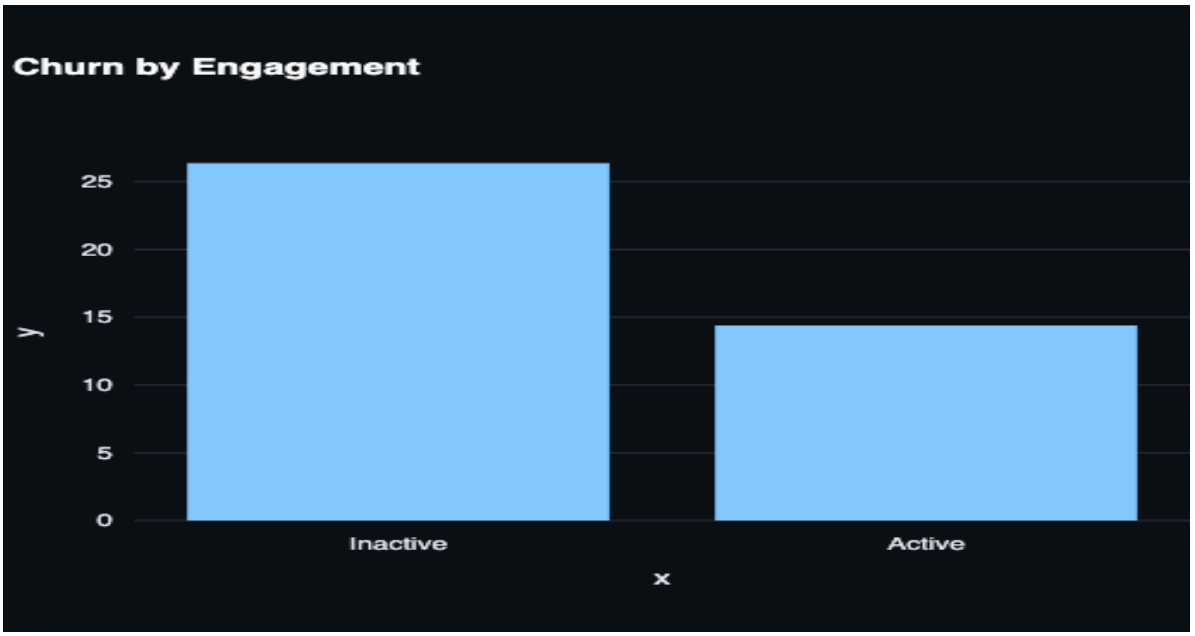
The dataset shows a significant proportion of customers who have exited the bank. This highlights how understanding churn drivers and implementing effective retention strategies plays a crucial role.



5.2.2 Engagement vs Churn

When we compare active and inactive customers, we get a clear image which shows that Inactive customers exhibit significantly higher churn rates,which means that engagement is one of the main causes of retention.

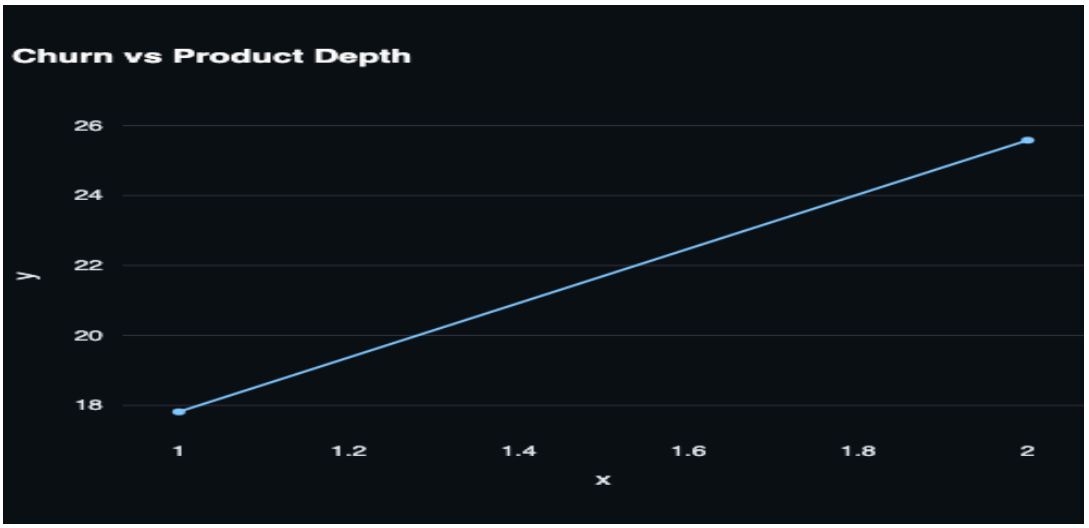
This means that the banks should focus on increasing customer interaction and activity.



5.2.3 Product Utilization vs Churn

The analysis shows that customers who have only one product are more likely to exit/churn. whereas, customers who are using multiple products are showing higher retention.

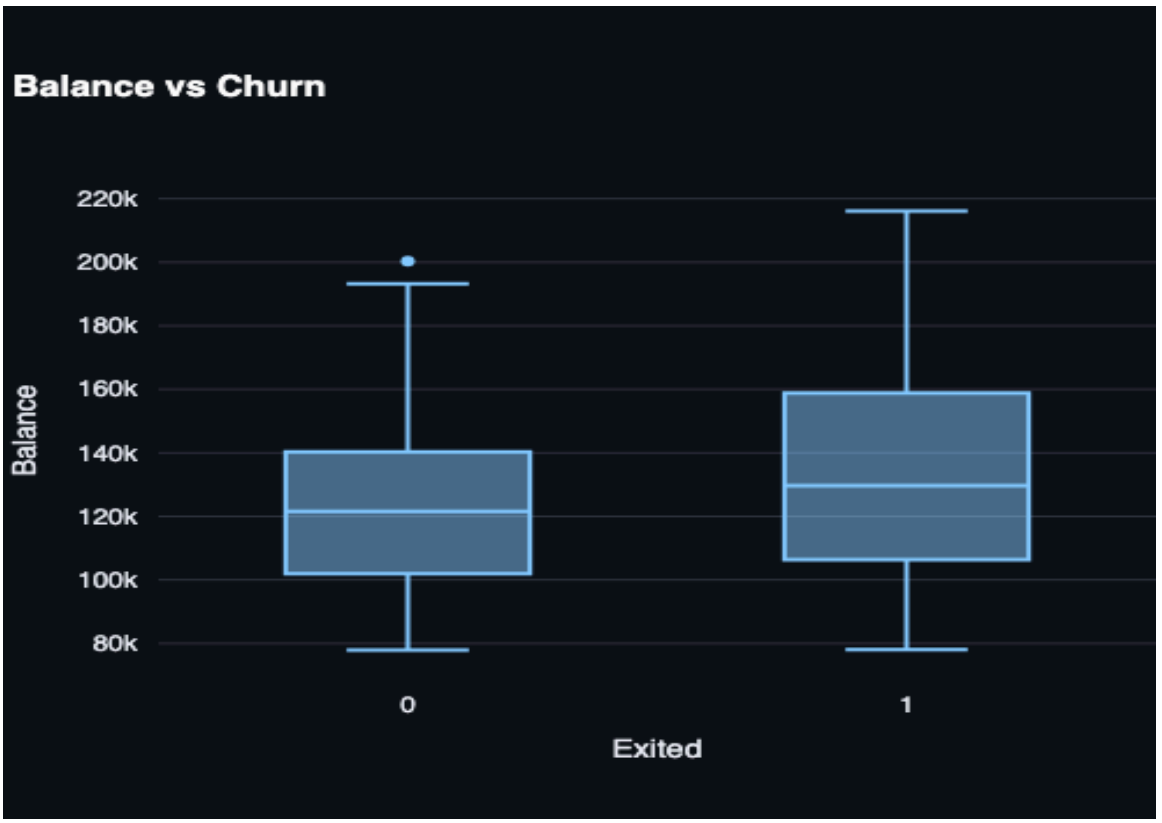
This tells that product diversification makes customer relationships strong and helps in reducing the churn probability.



5.2.4 Financial Behavior vs Churn

During study, it is found that the financial strength alone does not guarantee retention. Some customers with high account balances were found to be inactive and at risk of churn.

This shows how combining financial analysis with behavioral insights is important.



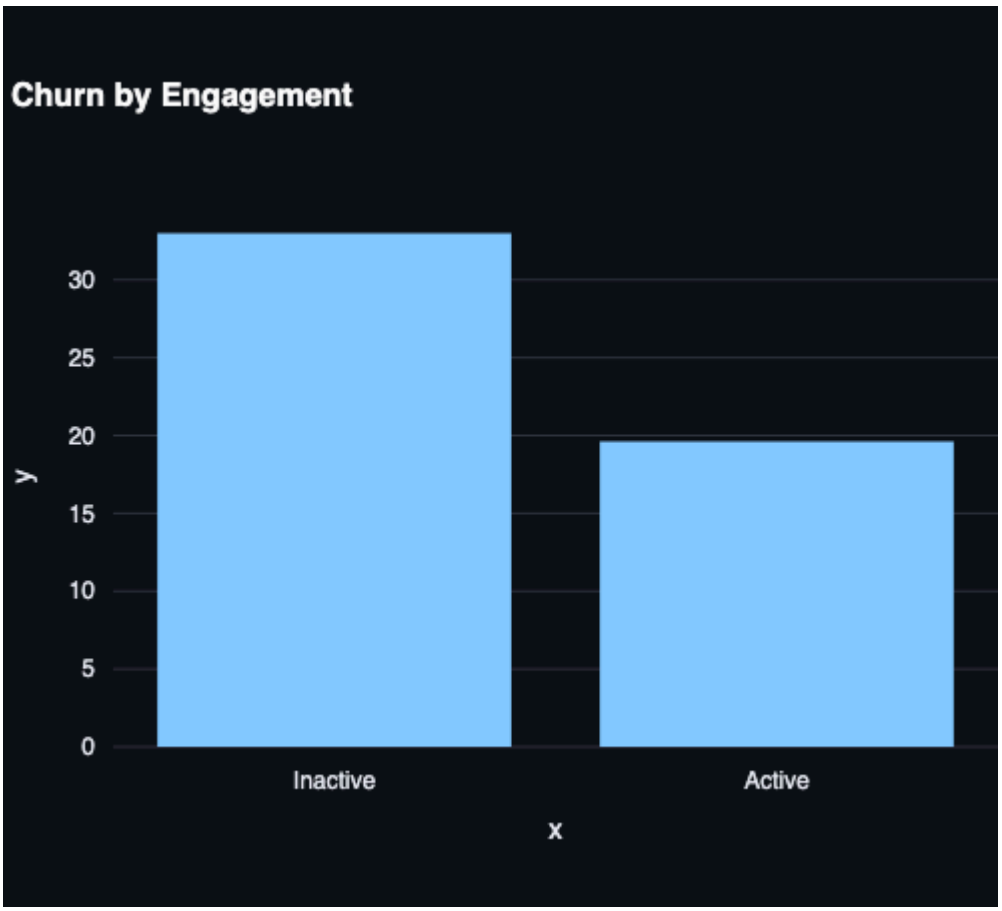
5.2.5 Demographic Insights

Analysis of the datasets shows that there are variations in churn across geography and gender, but these factors are less influential as compared to engagement and product usage.

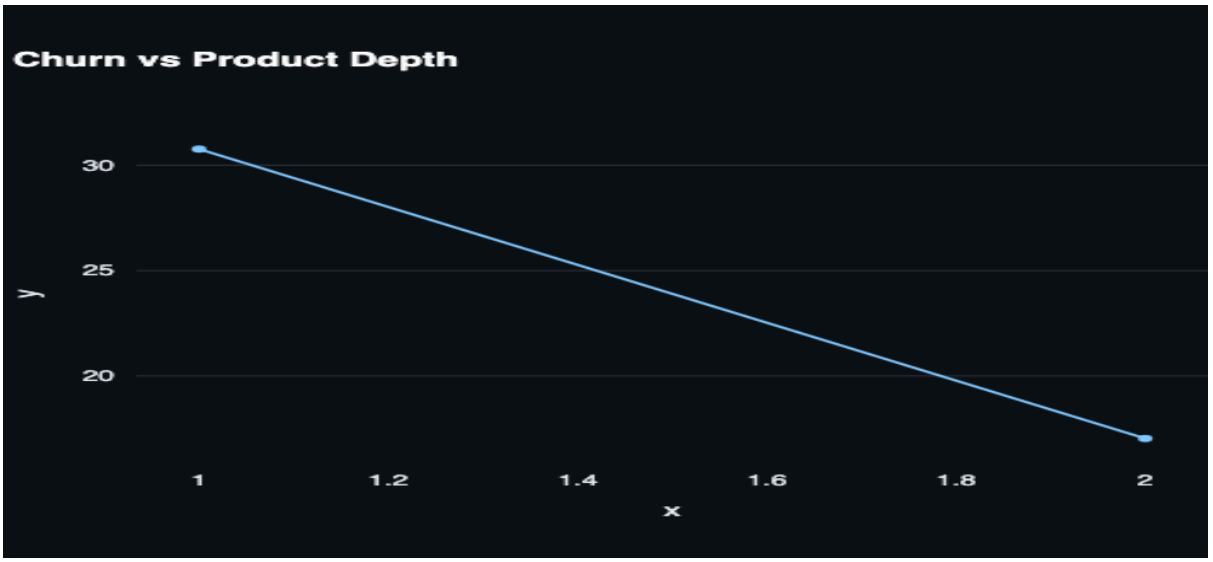
5.3 KPI Development

To quantify customer behavior and measure retention drivers, several key performance indicators (KPIs) were developed:

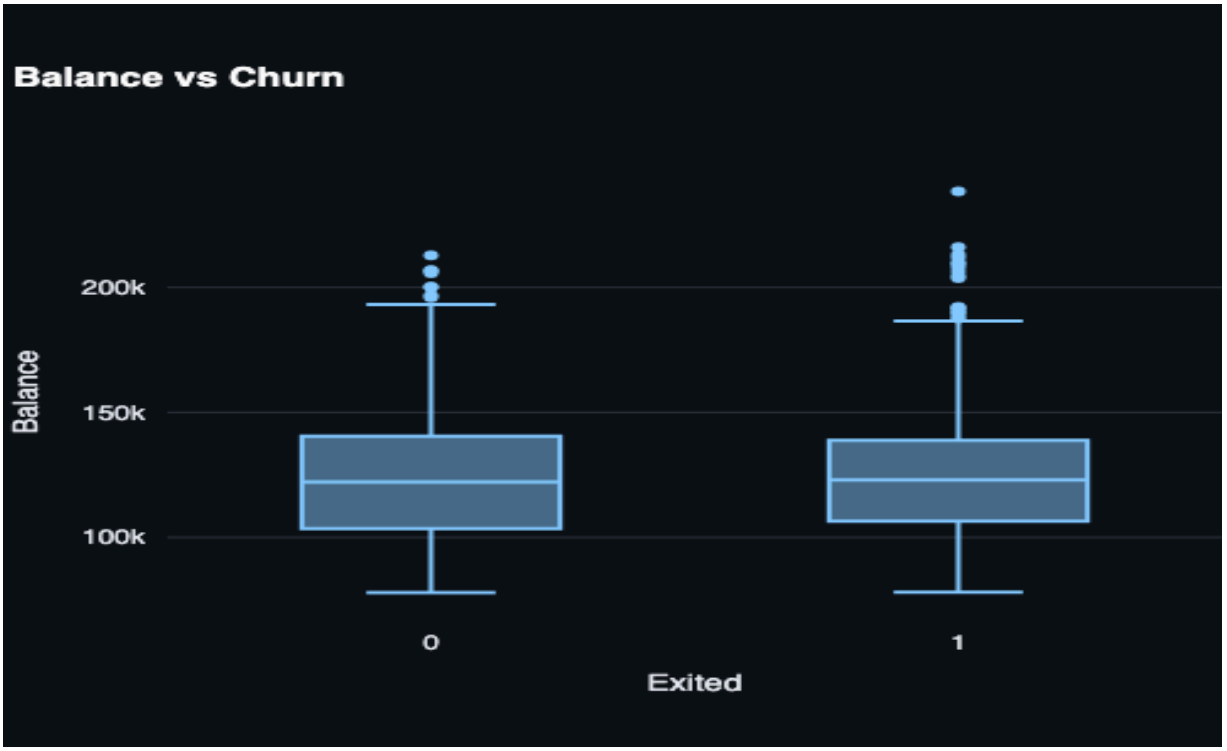
- **Engagement Retention Ratio:** it measures the difference in churn between active and inactive customers



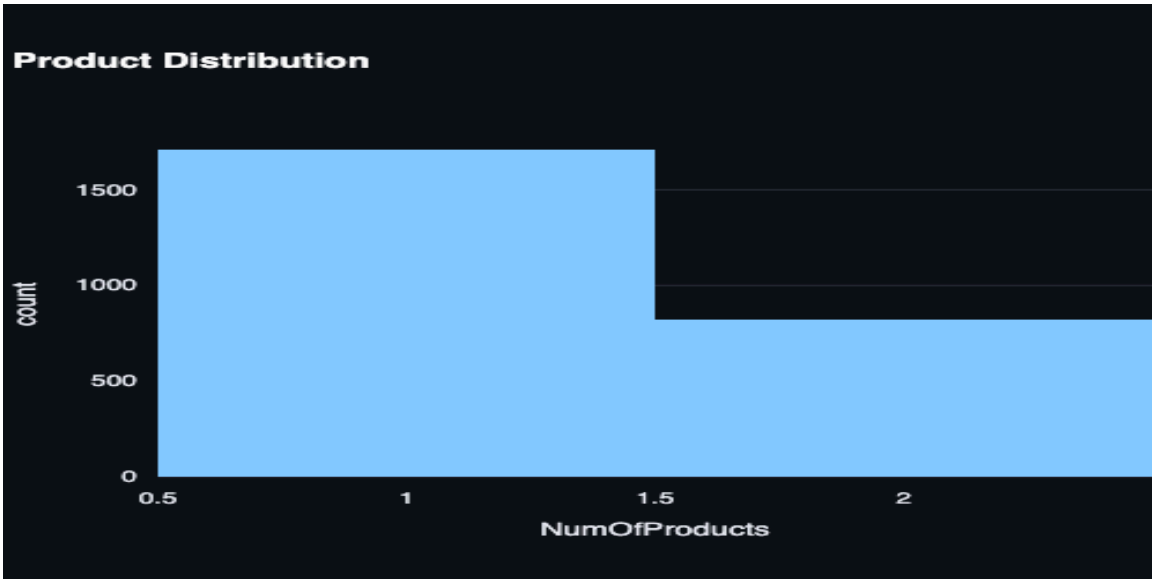
- **Product Depth Index:** it represents the average number of products used by customers



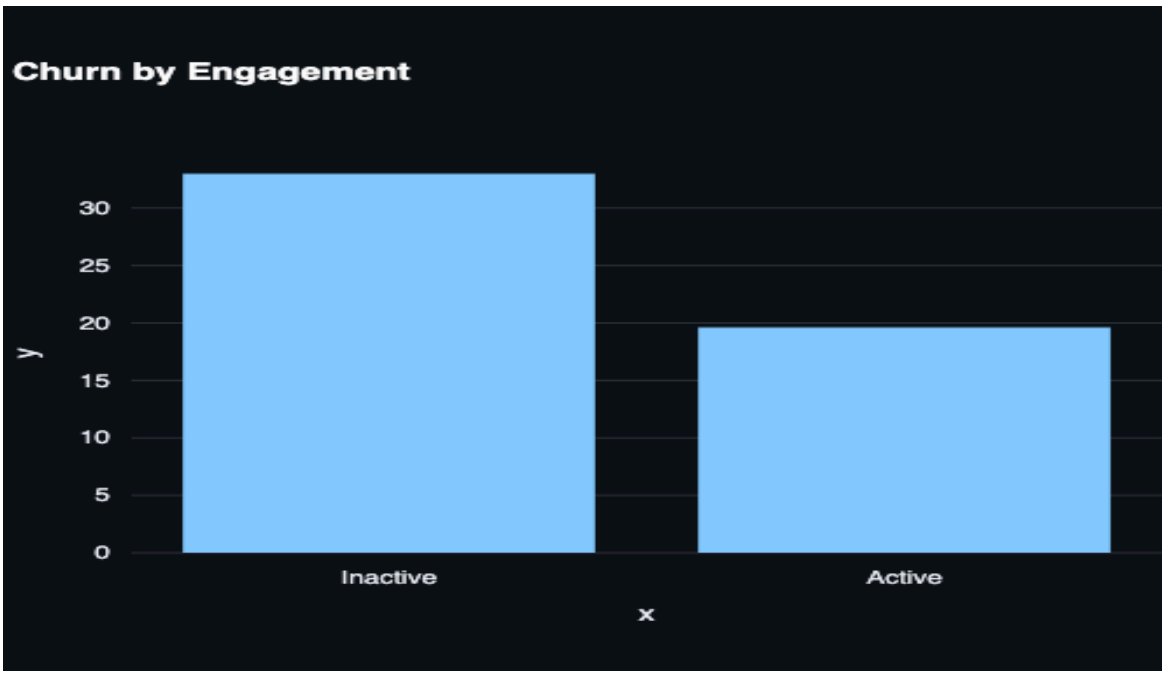
- **High-Balance Disengagement Rate:** it identifies high-value customers at risk due to inactivity



- **Credit Card Stickiness Score:** it evaluates the impact of credit card ownership on retention



Relationship Strength Index: it combines engagement and product usage to measure overall loyalty



The first important KPI used in this study is the **Engagement Retention Ratio**. This metric compares the churn rates of active and inactive customers. It makes it easier to understand how strongly customer engagement influences retention. A higher difference between these groups indicates that inactive customers are significantly more likely to churn.

Another key metric is the **Product Depth Index**, which represents the average number of products used by customers. This KPI helps in understanding whether customers who use more banking services tend to stay longer. It also provides insight into how product adoption contributes to building stronger customer relationships.

The **High-Balance Disengagement Rate** was developed to identify customers who have high account balances but are not actively engaged. These customers are financially valuable, but their low activity level makes them more likely to leave. This KPI is particularly useful for identifying hidden risks within the customer base.

The **Credit Card Stickiness Score** is used to evaluate whether customers who own a credit card are less likely to churn compared to those who do not. This helps in understanding the role of additional financial products in improving customer retention.

Finally, the **Relationship Strength Index** was created as a combined measure of customer engagement and product usage. By bringing these two factors together, this KPI provides an

overall view of how strong a customer's relationship with the bank is. Customers with higher relationship scores are generally more loyal and less likely to churn.

5.4 Dashboard Development

As part of this project, I developed an interactive dashboard using Streamlit to make the analysis more practical and easier to understand. The dashboard helps in presenting the data in a more visual and user-friendly manner so it can be easily understood.

The main objective of building this dashboard is to convert the analytical results into a format so that it can support real-time decision-making. Instead of doing only on static charts or tables, the dashboard allows users to interact with the data and explore different patterns more effectively.

The dashboard includes a set of key performance indicators (KPIs) that provide a quick overview of customer behavior by which we can see basic metrics such as total number of customers, churn rate, average balance, and average salary. There are also more advanced indicators like engagement retention ratio, product depth index, and relationship strength index were also included to give deeper insights into customer engagement and loyalty.

One of the most important features of the dashboard is the use of filters. Users can filter the data based on different factors such as geography, gender, number of products, and balance levels. This makes it possible to focus on specific customer segments and understand how churn patterns vary across different groups.

The dashboard also includes multiple visualizations to represent key relationships in the data. For example, it shows the difference in churn between active and inactive customers, the impact of product usage on retention, and the role of financial factors such as account balance. These visualizations help in identifying patterns and trends more clearly compared to raw data.

Another important aspect of the dashboard is the identification of high-risk customer segments. Customers who are with high account balances but low engagement are found with a potential risk group. This is useful for identifying customers who may appear valuable but are actually more likely to churn if no action is taken.

Dashboard not only supports the findings of the study but also provides a tool that can be used for continuous monitoring and better decision-making. By combining data visualization with interactivity, the dashboard makes it easier to translate analytical insights into real-world actions.

6. Key Insights

After analyzing the dataset and exploring different patterns, a few important observations were found that can help in understanding customer churn more clearly:

- ***Customer engagement makes a noticeable difference in retention***
  After analysis of data it is observed that customers who are inactive are much more likely to leave the bank. On the other hand, active customers tend to stay longer. This shows that staying connected with customers on a regular basis will influence retention.

- ***Using more products builds a stronger connection with the bank***
  Customers who are using only one product have chances to leave more often. In contrast, those who are using multiple services seem more comfortable continuing with the bank. This suggests that when customers are more involved with different products, their relationship becomes stronger.

- ***High balance does not always mean the customer will stay***
  One interesting thing I noticed is that even customers with high account balances are not always loyal. Some of them were not very active, which means they are likely to churn. So, financial value alone does not guarantee retention.

- ***There is a hidden risk among high-value but inactive customers***
  A specific group stood out during the analysis—customers who have high balances but low activity. These customers may look valuable at first, but they are actually at

risk because they are not actively engaged. This makes them an important segment to focus on.

- ***Stronger relationships reduce the chances of churn***
  Customers who are both active and using multiple products generally show better retention. It feels like when customers interact more and use more services, they develop a stronger connection with the bank, which reduces their chances of leaving.

- ***Customer behavior matters more than basic demographic factors***
  Although there are some differences based on location or gender, these factors do not seem to influence churn as much as engagement and product usage do. This shows that understanding how customers behave is more useful than just looking at who they are.

7. KPI Analysis

To make the analysis easier and identify the key trends in the data, I picked a couple of key performance indicators (KPIs) that simplified the analysis and allowed pointing to the significant trends.

Engagement Retention Ratio

This measure enabled me to make comparisons between the behavior of active and inactive customers. It was quite evident that the inactive customers are far more prone to being lost, which proves the point that engagement can be regarded as one of the most significant retention issues.

Product Depth Index

In the process of product usage analysis, I noted that the longer customers use more products. This KPI provides a rough concept of customer engagement with the bank, and the higher the values, the more likely, there are stronger relationships.

High-Balance Disengagement Rate

This was a more interesting metric. It assisted in identifying high balance customers who are not very active. These customers may appear worthwhile initially, but since they engage less, they are prone to churn.

Stickiness of Credit Card Score

It was this KPI that was used to observe whether having a credit card is any different. Based on the analysis, it appears that credit card customers are a bit more likely to remain, perhaps due to their being more entrenched in the bank services.

Relationship Strength Index

This is a combination of involvement and product consumption into a single measure. In my own experience, customers who score higher here are more likely to be loyal since they are active and consuming more than one product.

All in all these KPIs helped comprehend customer behavior in an organized manner. They also assisted in pointing out the obvious trends, rather than analyzing raw data, and made the insights more efficient.

8. Customer Segmentation

To make the analysis more practical, I divided customers into different groups based on their behavior. This made it easier to understand which customers are at risk and which ones are more stable.

- ***Active and Engaged Customers***
  These customers are regularly interacting with the bank and are generally less likely to churn. They represent a strong and stable segment.

- ***Inactive and Disengaged Customers***
  This group shows a higher tendency to leave. Since they are not actively using the bank's services, they are more likely to disconnect over time.

- ***Active Customers with Low Product Usage***
  These customers are active but only use one or very few products. They are not at immediate risk, but they present a good opportunity for cross-selling to strengthen their relationship with the bank.

- ***High-Balance Inactive Customers***
  This is a very important segment identified during the analysis. These customers have high financial value but low engagement. Even though they seem valuable, they are actually at risk of churn and need special attention.

By dividing customers into these segments, it becomes much easier to design targeted strategies. Instead of applying the same approach to everyone, banks can focus on each group based on their behavior and needs.

9. Recommendations

Based on the analysis and insights from this project, I came up with a few practical suggestions that banks can use to improve customer retention and build stronger relationships.

- ***Focus more on improving customer engagement***
  From the analysis, it is clear that inactive customers are more likely to leave. Because of this, banks should try to increase customer interaction through regular communication, personalized messages, and better digital engagement. Even small improvements in activity levels can make a difference.

- ***Encourage customers to use multiple products***
  Customers who use only one product seem more likely to churn. So, banks should promote cross-selling by offering bundled services or personalized product recommendations. This can help customers become more connected with the bank.

- ***Pay special attention to high-value inactive customers***
  One of the most important findings was the presence of customers who have high balances but low engagement. These customers may look valuable, but they are actually at risk. Banks should identify them early and reach out with personalized offers or relationship management support.

- ***Introduce loyalty and reward programs***
  Rewarding customers for staying active and using multiple products can help strengthen long-term relationships. Simple incentives, cashback offers, or exclusive benefits can encourage customers to remain engaged.

- ***Use data regularly for monitoring and decision-making***
  Instead of relying only on periodic analysis, banks should continuously track

customer behavior using dashboards and KPIs. This will help in identifying problems early and taking timely action before customers decide to leave.

Overall, these recommendations are based on the patterns observed in the data and are focused on practical actions that can be implemented in a real-world banking environment.

10. Conclusion

This project enabled me to realize that it does not only depend on the financial aspect to retain the customers in the banking industry, but rather the way the customers relate to the bank. During the analysis, it became obvious that the engagement and product usage is a significantly larger factor than I originally thought.

Among the main lessons is that dormant customers are far more susceptible to churning. This indicates that customers are not just any assets that the banks must have, but they must maintain the clientele in a constant activity. In the same manner, customers who consume a number of products would be more likely to have longer stay hence implying that better relationships are established as customers become more engaged with services provided by the bank.

The other significant thing noticed is that the high-value customers do not always remain loyal. It was identified that some customers with high balances are not active and this would form a potential risk segment. It reveals the need to consider more than financial data and concentrate more on how customers behave.

On the whole, this project allowed me to realize how data may be utilized to find patterns and make more effective decisions. With the help of insights based on engagement, product usage, and customer segmentation, developing more efficient retention strategies becomes a possibility.

Finally, customer behavior could assist banks in transitioning to a more proactive and data-driven retention strategy as opposed to relying on financial indicators. This can eventually result in more customer relations and better long term results.